

Question:
Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Reasoning + Answer:

Let's solve this step by step.

- ### Step 1: Understand the information
- Josh buys a house for \$80,000.
 - He spends \$50,000 on repairs.
 - The repairs increase the value of the house by 150%.

We are to find the profit he made.

Step 2: What does "increased value by 150%" mean?

"150% increase" means the new value of the house is the original value plus 150% of the original value.

Let's compute the new value of the house after the repair:

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$$\text{Original value} = \$80,000$$

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$$\text{150\% increase} = 1.5 \times 80,000 = 120,000$$

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So, the new value of the house = \$80,000 + 150% of \$80,000 = \$120,000

Step 3: Calculate total cost

- Josh spent:
- \$80,000 to buy the house
 - \$50,000 on repairs

Total cost = \$80,000 + \$50,000 = \$130,000

Step 4: Profit = New Value - Total Cost

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$$\text{Profit} = 120,000 - 130,000 = -10,000$$

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Negative profit means loss, not profit.

Wait! this is a contradiction? Did we misread?

Let's double-check:

> "This increased the value of the house by 150%"

That could mean either:

1. 150% increase on original price! we did that.
2. Total value became 150% of original value? That would be different.

"Increased by 150%" usually means a 150% increase, not 150% of the original value.

But if the original price was \$80,000, then 150% increase gives:

- 150% of \$80,000 = \$120,000
new value = \$80,000 + \$120,000? No, only 150% increase.

Actually:

$$\begin{aligned} \text{New value} &= \text{original} + 150\% \text{ of original} = 80,000 + 150\% \text{ of } 80,000 \\ &= 80,000 + (1.5 \times 80,000) = 80,000 + 120,000 = \$200,000 \end{aligned}$$

No! that's not possible.

Actually, 150% increase on \$80,000:

- 100% of 80,000 = 80,000
- 150% of 80,000 = 1.5 × 80,000 = 120,000
- So, new value = 80,000 + 120,000 = \$200,000?

No! Wait, 150% of \$80,000 is 120,000.
So, increase of 150% new value = 80,000 + 120,000 = \$200,000

Yes.

So total